

# Biophysics of radiations

## INTRODUCTION

on living matter is often perceived as a harmful effect. The scientific study of matter-radiation interaction provides the possibility of correctly understanding numerous and complex experimental facts; this is radiobiology.

Specialist scientists or doctors draw from these studies a coherent body of knowledge oriented towards radiation protection.

Radiation protection comes from the desire to avoid pathology induced by radiation on a sick or healthy population, but also on the descendants of these populations.

The applications of radioactivity, particularly industrial applications, have generated sometimes unreasonable fears. The pathology induced by medical applications required the implementation of radiation protection standards in 1928.

Cancer patients are treated with radiation, this is radiotherapy. The effect is no longer harmful, on the contrary it is beneficial for the patient.

All radiation, ionizing or non-ionizing, presents biological effects, useful or harmful to life. Therefore, the current importance of radiation and the prospects for its development require doctors and biologists to know the principles of their uses. In this course, we present, in the first part, the general physics of radiation. In the second part, we are interested in the study of the biological effect of radiation and their applications in the medical field.

## Part 1: GENERAL PHYSICS OF RADIATION

Radiation is a mode of energy propagation through space, in the form of waves or particles. Radiation can only be detected through its interactions with matter: it releases all or part of its energy and undergoes changes in its behaviour

### I . Classification of radiations

- Depending on the nature of the radiation, they are classified as electromagnetic radiation and particulate radiation.
- Based on their effects on biological matter, they are classified as ionizing radiation and non-ionizing radiation.

#### I-1 Electromagnetic Radiation

##### 1-1-1 Wave-like Nature of RME

Electromagnetic radiation is emitted by the vibrations of the atomic electrons that compose it (with two components: electric and magnetic).

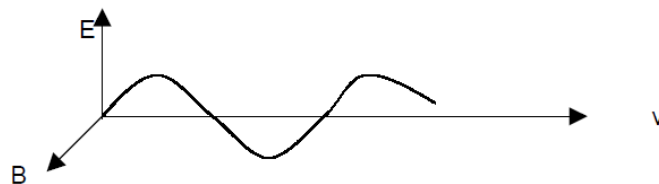
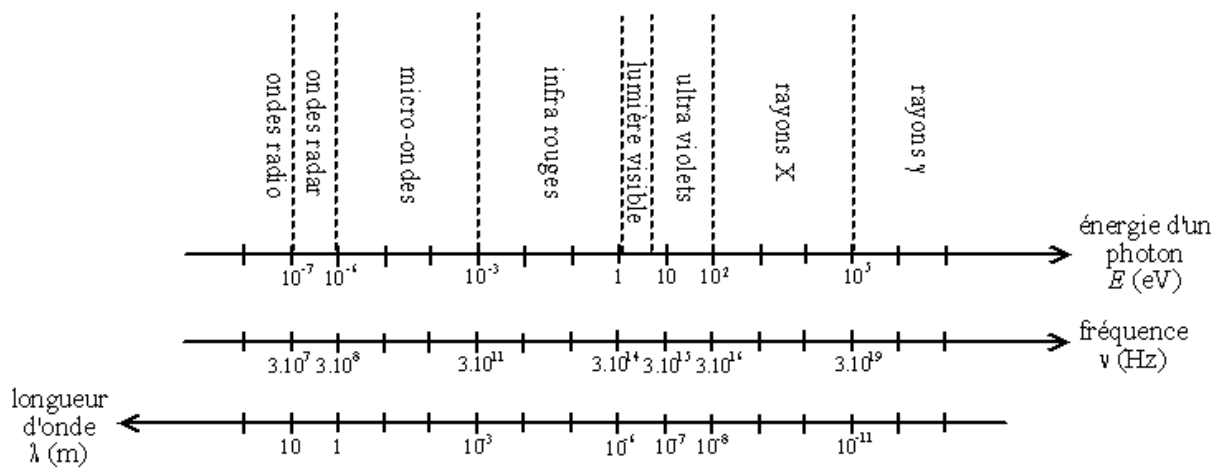


Fig. 1: Electromagnetic radiation

These radiations are characterized by their speed in a vacuum (celerity)  $\approx 3 \times 10^8$  m/s and their frequency ( $\nu = 1/T$ ), where  $T$  is the period or wavelength ( $\lambda = CT = C/\nu$ ). They are sometimes distinguished by their wavenumber ( $\sigma = 1/\lambda$ ) or angular frequency ( $\omega = 2\pi/T$ ).

Based on their frequencies, they are classified according to the energy of their photons, or, equivalently, their wavelength or frequency – the '**electromagnetic spectrum**'.



**Fig. 2** Classification of electromagnetic radiation by energy, frequency and wavelength.

## I-1-2 Corpuscular aspect of REM

The wave nature of electromagnetic waves (EMW) is not sufficient to explain certain phenomena such as the photoelectric effect. This leads to the introduction of the corpuscular nature of photons in conjunction with their wave nature.

### I-1-2-1 Einstein's hypothesis on the quantization of radiation

It was in 1905 that Einstein proposed his hypothesis on the quantization of radiation. The emission and absorption of radiation of frequency  $\nu$  occurs through "quanta" or "photons" of energy  $E = h\nu$ .

The photon remains localized in space (it does not have infinite extension); once emitted, it moves away from the source at the speed of light.

$h$  is the constant introduced by Planck in 1900:  $h = 6.63 \times 10^{-34}$  Joule-s

in summary, the photon behaves as both a particle and a wave. In 1923, Louis de Broglie partially solved the problem of the wave-particle duality of electromagnetic waves. This allows us to apply Einstein's relation  $E = mc^2$

$$\lambda = h/mc \Rightarrow E = h\nu = hc/\lambda = mc^2$$

The energy of the photon associated with the wavelength in a vacuum will then be:

$$h\nu(J) = (3 \times 10^8 \times 6 \times 10^{-34})/\lambda(m) = (1 \times 8 \times 10^{-25})/\lambda(m)$$

### I-1-2-2 Wave-like aspect of matter

De Broglie suggests generalizing this duality to matter (material particles): "Every moving particle, whatever its nature, must always be considered as associated with a wave: the particle's motion is deduced from the propagation laws of the associated wave."

Wave-particle duality leads to associating a wave of wavelength  $\lambda$  with every particle, material or intangible, with momentum of value  $p$  such that:

$$\lambda = h/mv = h/p$$

According to the de Broglie relation, for the wave-like behavior of matter to manifest itself, the particle's mass must be relatively small. This is the case for microscopic particles such as the electron, proton, and neutron.

Otherwise, the wavelength of the associated wave is so short that there is no obstacle or opening small enough to diffract the wave. Therefore, the wave-like behavior of this particle of matter cannot manifest itself.

## I-2 Particulate radiation

We will group under the term corpuscular or particle radiation radiation made up of elementary particles of non-zero rest mass. In the medical field, this will essentially be radiation made up of electrons, helium nuclei ( $\alpha$  particles), protons, and more rarely neutrons. These particles can be produced during radioactive decays or by means of accelerators. In the medical field, this radiation always has energies largely capable of ionizing atoms. In practice we distinguish particles according to their charge: charged particles or uncharged particles (neutral).

### I-2-1 Charged particles

These will essentially be relatively heavy charged particles and lighter charged particles. These radiations act with the electrons of the target material

Light charged particles (electrons): electrons,  $\beta^-$  and  $\beta^+$  radiation, internal conversion electrons, Auger electrons...All these particles differ only in their mode of production and are fundamentally identical in nature.

The generic term electron will be applied to these particles.

- Heavy charged particles: protons,  $\alpha$  particles, heavy ions, nuclear fission residues...The behavior in matter of these entities is subject to the same laws as light charged particles, however the effect of mass intervenes, which has the advantage of simplifying the mathematical and physical study.

### I-2-2 Uncharged particles (neutrons)

This is radiation which acts on the nuclei of the target matter. In the medical field, we only consider neutron radiation. A neutron beam will be an excellent way to incorporate neutrons into nuclei or to observe collisions with them. Given the very limited volume of the nucleus within the atom, these shocks will however be infrequent, which explains why a beam of neutrons will be able to penetrate deep into an organism. The neutron-matter interaction could, however, increase significantly. If the de Broglie wavelength associated with neutrons is of the order of the dimensions of the atom (10–10 m). This will occur for relatively low energies, below 1eV. Such neutrons will be qualified as thermal neutrons, because they have kinetic energies of the order of that given by thermal agitation. Neutrons larger than 0.8 MeV are called fast neutrons. Between 1 eV and 0.8 MeV, we will speak of epithermal neutrons.

Clinical trials are underway to evaluate the value of thermal neutrons in the treatment of certain cancers.

## II- X-ray

X-rays were discovered in 1895 by W. Röntgen in Würzburg, Germany. He named the rays he discovered X-rays with the X as the unknown in Mathematics.

X-rays are electromagnetic radiation, which occupy the range from 0.03 nm to 10 nm. Although there are no clearly defined limits.

### II-1 Production of X-rays

One application of electron interactions with matter is the production of X-rays. The X-ray beams used in medicine are generally produced by X-ray tubes called Coolidge tubes, the principle of which is shown in Figure 8.

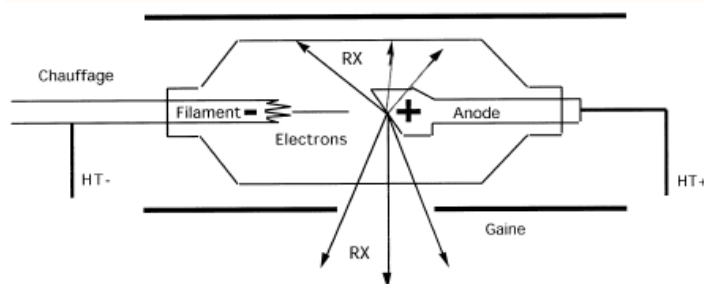


Fig.4: Diagram of a Coolidge X-ray tube

In a Coolidge X-ray tube, dedicated for imaging, the emission of electrons is obtained by thermoelectric effect, using a filament heated to a sufficient temperature by a current  $i$ . The electrons leave the filament under the effect of an intense electric field which accelerates them to the target. This electric field is obtained using a potential difference  $U$  whose value is usually between 30 and 150 kV for radiodiagnostic tubes, and a few hundred kV for tubes used in radiotherapy. The emitting filament constitutes the negative pole or cathode, the metallic target constitutes the positive pole or anode. These electrodes are placed in a glass enclosure where a high vacuum is created. The tube is surrounded by lead shielding pierced by a window through which the RXs emerge.

## II-2 X-ray spectrum

The emission of X-rays is the consequence of the interactions that occur between fast electrons and the particles of the metal into which they penetrate. These fast electrons have a kinetic energy  $E_0 = e.U$

### II-2-1 Continuous braking spectrum

When an electron with kinetic energy  $E_0$  approaches the nucleus of positive charge, it radiates energy in the form of an electromagnetic wave. This energy is taken from the kinetic energy of the electron in the form of RX. The electron therefore loses energy and will have a final energy  $E_c$ . We say that the electron is slowed down. The energy of the X-ray photons emitted is:

$$E = E_0 - E_c = h\nu$$

Three possibilities may arise:

- The electron undergoes complete braking  $h\nu = E_0 = e.U$

Thus the maximum energy of the RX is numerically equal to  $U$

- the incident electron loses part of its energy.  $0 < h\nu < E_0$

- The electron passes without any loss of energy: no X-ray photon emitted.

The total energy emitted in the form of low energy photons is greater than the total energy emitted in the form of high energy photons.

## III- Interactions of radiation with matter

Ionizing radiation has in common the property of causing ionizations and excitations to appear in the material environments where they penetrate. We distinguish between directly ionizing radiation which is formed by charged particles ( $\alpha$ ,  $\beta$ , etc.) and indirectly ionizing radiation (RX,  $\gamma$  radiation, neutrons). Ionizations are at the origin of the effects used for the detection of radiation, as well as at the origin of the biological effects observed on biological environments. In the field that interests us, electrons have a major importance.

### III-1 Interactions of photons with matter

When a mono-energetic beam of photons passes through a material, it interacts, in a random manner, with the atoms of this material and emerges attenuated. This attenuation is due to both diffusion, that is to say a change of direction of the incident photons, and absorption, that is to say a transfer of the energy of the incident photons to the material crossed.

#### III-1-1 Law of exponential attenuation:

Let  $N_0$  be the total number of photons arriving on the screen per unit area. Let  $N(x)$  be the number of photons having passed through a thickness  $x$  of the screen without being attenuated, that is to say neither reabsorbed nor diffused, we call them transmitted photons.

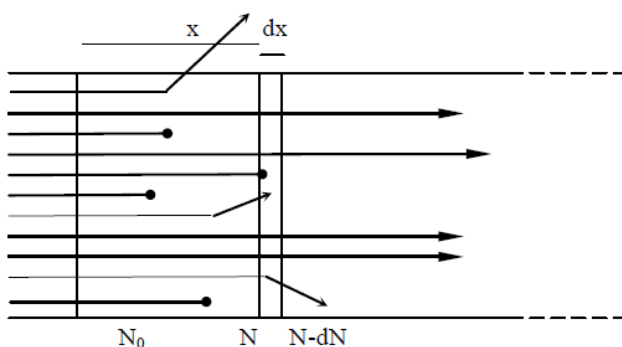


Fig. 5: Attenuation of photons by a screen of thickness  $x$

The number of photons,  $dN$ , attenuated by a tiny thickness  $dx$  is such that:

$$dN = -\mu N dx$$

$$\Rightarrow N = N_0 e^{-\mu x} \Leftrightarrow \text{Log } N = -\mu x + \text{Log } N_0$$

$\mu$  is the linear attenuation coefficient, or: probability of interaction per unit of length, its unit is  $\text{cm}^{-1}$  ( $\mu$  depends on the nature of the material and the energy of the photon). We often use mass attenuation coefficient  $\mu/\rho$  ( $\text{cm}^2.\text{g}^{-1}$  or in  $\text{m}^2.\text{kg}^{-1}$ ) which is independent of the physical state of the material.

Thus, we can, by interposing a screen, attenuate a beam of photons but never stop it completely.

Since the photons considered are monochromatic, an analogous relationship connects the incident energy ( $I_0$ ) of the beam and its energy after crossing a thickness  $x$ :  $I_x = I_0 \cdot e^{-(\mu x)}$

If the beam passes through several media with different attenuation coefficients:  $\mu_1, \mu_2, \mu_3, \dots, \mu_i$ , over thicknesses  $x_1, x_2, x_3, \dots, x_i$ , the number of emerging rays will be:

$$N(x) = N_0 \cdot e^{-(\mu_1 x_1 + \mu_2 x_2 + \mu_3 x_3 + \dots + \mu_i x_i)}$$

### III-1-2 CDA

We call Half Attenuation Layer, CDA, the thickness which halves the initial intensity  $I_0$

Thus, if we interpose a screen with a thickness of 10 CDA, we will have an attenuation of 99.9% of photons.

### II-1-3 Elementary interactions

The attenuation of the beam of X and  $\gamma$  photons occurs according to several mechanisms. Only three types of interaction are of interest in the medical application. They are: the photoelectric effect, the Compton effect, the Compton effect and the creation of pairs.

#### III-1-3-1 Photoelectric Effect

The photoelectric effect results from the transfer of all of the energy of the incident photon onto a bound electron in the target's electron procession. This is a total absorption effect. This effect only occurs if the energy  $E$  of the photon is greater than the binding energy  $W_l$  of the electron. The electron is then ejected from the electron procession with a kinetic energy  $E_c$  such that:

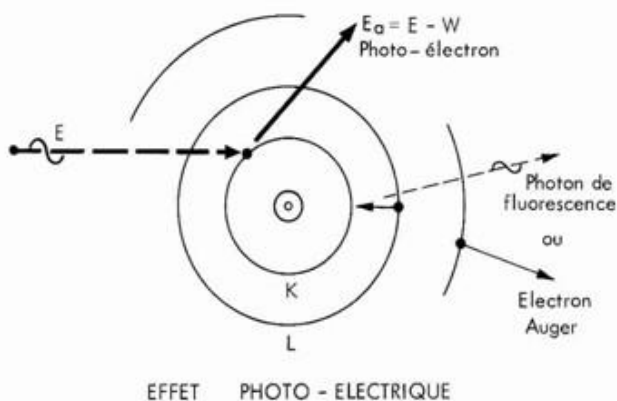
$$E_c = E - W_l$$

The expelled electron exhausts its energy in ionization and excitation. The energy  $E_c$  is therefore completely absorbed in the target. It leaves a vacant place and there is consequently an electronic rearrangement. This vacant place will be filled by an electron from a more external shell. This replacement is accompanied by a release of energy which can be:

- Either diffused in the form of characteristic X-ray photons or fluorescence photons.

Predominant phenomenon for high  $Z$  materials, such as lead.

- Or diffused in the form of Auger electrons which will then in turn be absorbed secondarily.



**Fig. 6:** Photoelectric effect and electronic rearrangement

Attenuation coefficient:  $\tau$

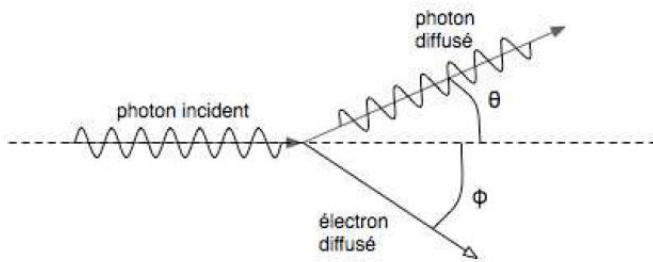
We define the linear attenuation coefficient  $\tau$  linked to the photoelectric effect. We show that, for a screen of density  $\rho$  and atomic number  $Z$ , the mass attenuation coefficient  $\tau/\rho$  of photons of energy  $E$  verifies the approximate relationship of Bragg and Pierre:

$$\tau/\rho \propto k \cdot Z^3/E^3 \text{ (k is a constant that does not depend on the material)}$$

The photoelectric effect is important especially for heavy elements and low energy photons.

### III-1-3-2 Compton Effect

The incident photon interacts with a free or weakly bound electron of an atom, gives it part of its energy and diffuses in a direction  $\theta$ , the electron, called the Compton electron, is ejected forward. The energy of the incident photon is shared between the incident photon and the Compton electron.



**Fig.7:** Collision of a photon with an electron at rest.

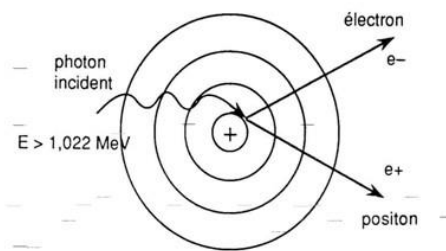
#### Attenuation coefficient: $\sigma$

We define a linear attenuation coefficient of the Compton effect:  $\sigma$  and the mass attenuation coefficient:  $\sigma/\rho$ . It is shown that  $\sigma/\rho$  is practically independent of the target material and decreases slowly when the energy  $E$  of the incident photon increases.

### III-1-3-3 Materialization Effect

The process occurs for very energetic photons ( $> 2m_0c^2=1.022$  MeV) passing close to a nucleus: the incident photon of energy  $E$  materializes in the form of an electron ( $e^-$ ) and a positron ( $e^+$ ) of the same mass  $m_0$  and respective kinetic energy  $E_{ce1}$  and  $E_{ce2}$ .

The energy equation is:  $E = 2m_0 c^2 + E_{ce1} + E_{ce2}$



**Fig 8:** Pair creation phenomenon

#### Attenuation coefficient: $\pi$

The linear attenuation coefficient linked to pair creation,  $\pi$ , increases approximately as the atomic number  $Z$ . It is zero for values of  $E < 1.022$  MeV otherwise increases with  $E$ .

### III-1-4 Relative importance of elementary interactions

The overall attenuation coefficient  $\mu$  results from these three effects. The photons which pass through the material without interaction are those which have not been the object of any of these interactions.

And we have:  $N(x)=N_0 e^{-(\mu x)}=N_0 e^{-(\tau x)} e^{-(\sigma x)} e^{-(\pi x)}$

So  $\mu = \tau + \sigma + \pi$  and  $\mu/\rho = \tau/\rho + \sigma/\rho + \pi/\rho$

The relative importance of the three effects depends fundamentally on the energy of the incident photon. We can divide the spectrum into three areas where each of the effects predominates:

- On the low energy side ( $< 100$  keV) the Photoelectric effect is largely predominant over the Compton effect.
- At medium energies (100 to a few MeV), the Compton effect is predominant
- At high energies ( $> 5$  MeV), materialization is the predominant effect.

The figure above illustrates the importance of the use of lead in medicine as a protective material since it absorbs by photoelectric effect most photons whose  $E < 1$  MeV.

We can thus decompose the attenuation coefficient  $\mu$  into a part  $\mu_d$  corresponding to the energy diffused and a part  $\mu_t$  corresponding to the energy transferred to the material (or absorbed by it).

- In medicine, we do not use incident photons of more than 1.3 MeV

- In radiodiagnosis, the photons are between 60 and 140 keV so that we have both a photoelectric effect and a Compton effect as a function of Z, thus allowing contrast between tissues of different densities. The overly predominant Compton effect is a source of “blur” so we try to limit them.
- For radiotherapy, photons are between 200 and 250 keV.

## **II-2 Interactions of charged particles with matter**

### **III-2-1 General case: any charged particle**

This involves obligatory interaction with the electrons of the target's atoms or with its nuclei. The interaction process with electrons is dominant up to energies of 100 MeV in water and 10 MeV in lead.

#### **III-2-1-1 Interaction with an electron of the target atom**

During this interaction, incorrectly called collision, the energy  $\Delta E$  given up by the incident particle is transferred to an electron of the target atom. Let  $W_l$  be the binding energy of this electron. Two phenomena can occur depending on whether or not  $\Delta E$  is sufficient to eject the electron from its orbit:

- Ionization: if  $\Delta E \geq W_l$ , the electron is ejected from its orbit with kinetic energy ( $W_l - \Delta E$ ) and ionization of the target atom occurs. This creates a pair of ions formed by the positive ion which has lost an electron and the negative ion which has gained an electron.
- Excitation: if  $\Delta E < W_l$ , the energy transfer  $\Delta E$  cannot produce any ionization but takes the electron to a higher energy level, with excitation of the target atom. The transferred energy is secondarily dissipated in the form of a low-energy electromagnetic emission.

Following the excitation and ionization of the atom, a rearrangement of the procession occurs.

### **Linear Energy Transfer (LET)**

We call Linear Energy Transfer or TLE, the quantity of energy transferred to the target medium by the incident particle per unit of trajectory length. The TLE is expressed in keV. $\mu\text{m}^{-1}$ . The TLE is given by the approximate formula:

#### **Linear Ionization Density (LID)**

We call Linear Energy Density DLI the number of ion pairs created by the incident particles, per unit of trajectory length. The DLI is expressed in ion pairs. $\mu\text{m}^{-1}$ . If  $W_i$  is the average energy transferred for each ionization, we have:

$$DLI = TLE / W_i$$

#### **III-2-1-2 Interaction with the nucleus of the target atom**

When charged particles pass near the nucleus of an atom of the target, their trajectory is deviated. This leads to a loss of kinetic energy, emitted in the form of RX called braking radiation.

### **III-2-2 Case of light charged particles: electrons (negaton and positrons)**

Fast moving electrons are obtained by:

- the  $\beta$  emission of radioelements
- electron accelerators
- the projection of secondary electrons by X or  $\gamma$  photons.

Their kinetic energy,  $E_c$ , is between a few keV and a few MeV

#### **III-2-2-1 the loss of energy of light particles**

The interaction process with electrons is dominant up to energies of around 100 MeV for water and 10 MeV for lead. In water, the TLE is approximately 0.25 keV. $\mu\text{m}^{-1}$  and DLI is approximately 8 ion pairs. $\mu\text{m}^{-1}$

The particle trajectories are broken lines. In water the sum of the lengths of all the segments of the trajectory is given by the approximate formula:

$$\text{length (cm)} = \text{initial energy of the particle in MeV} / 2$$

The distance which separates the entry point of the particle into the target medium from the end point of its trajectory is called average penetration depth. The maximum value of this distance  $P_{\text{max}}$ , expressed in cm, is given by the approximate formula:

$$P_{\text{max}} = 0.215 / \rho \left[ E(\text{MeV}) \right]^{1.66} \text{ where } \rho \text{ is the density of the material}$$

$P_{\text{max}}$  is of the order of a few cm in air and a few mm in water for energies used in medicine.

The positron has the same type and characteristics of interaction with matter. The main difference between electron and positron is the annihilation reaction for the latter.

### III-2-3 Case of heavy particles ( $\alpha$ , p)

For these very heavy particles (the mass of a proton is approximately 1840 times greater than that of an electron) the energy transfers are significant and the trajectory is approximately rectilinear.

In water, the TLE is approximately  $150 \text{ keV} \cdot \mu\text{m}^{-1}$ , DLI of  $4500 \text{ ion pairs} \cdot \mu\text{m}^{-1}$ . The length of the trajectories, is almost equal to the penetration depth, is given by the approximate formula:

$$L(\text{cm}) = (E(\text{MeV})) / 1500$$

This depth does not exceed a few cm in air and a few tens of  $\mu\text{m}$  in water. These particles are completely stopped by a simple sheet of paper or by the stratum corneum of the skin. They are therefore safe in the event of external irradiation but are dangerous if internal irradiation.

### III-3 Interactions of neutrons with matter

Uncharged particles, neutrons have negligible interaction with electrons. Their kinetic energy is absorbed by their interaction with the nuclei until they come to a complete stop. The probability of interaction with nuclei is very low due to the diameter of the atom. As a result, neutrons are very penetrating. A neutron which has a kinetic energy greater than 1 keV is said to be fast; its slowing down occurs by collision with the nuclei. The loss of kinetic energy is maximum for a hydrogen nucleus. So water is a relatively very attenuating material for fast neutrons. For slow neutrons, the interaction with the nuclei results in the absorption of the neutron. This phenomenon is widely used in the nuclear industry.

## Part 2: BIOLOGICAL EFFECTS OF RADIATION AND APPLICATIONS IN MEDICINE

Living matter (virus, cell, tissue or animal) is composed of atoms structured into molecules. Subjected to irradiation by ionizing or non-ionizing radiation, this material receives energy, with or without transfer of matter.

Ionizing, as well as non-ionizing, radiation can have deleterious effects, harmful to health, or on the contrary useful and even necessary for life.

### I-IONIZING RADIATION

Ionizing radiation is radiation capable of depositing enough energy in the matter it passes through to create an ionization of atoms of molecules of biological interest, rich in nitrogen, oxygen and hydrogen, whose ionization energies are of the order of ten electronvolts.

| Atom                   | N     | O     | H     | C     | Na   | K    |
|------------------------|-------|-------|-------|-------|------|------|
| Ionization energy (eV) | 14,51 | 13,57 | 13,54 | 11,24 | 5,13 | 4,10 |

In general, a threshold of 13.6 eV is used for ionizing radiation. These ionizing radiations, when controlled, have many beneficial practical uses (health, industry, etc.) But for living organisms, they are potentially harmful in the long run and fatal in the event of high doses.

The most energetic radiation transfers enough energy to the electrons of matter to tear them from their atom. The atoms thus deprived of some of their electrons are then positively charged. Neighboring atoms that host the electrons become negatively charged.

Positively or negatively charged atoms are called ions. Atoms that lost at least one electron became positive ions (cations), while atoms that received at least one electron became negative ions (anions). Radiation capable of causing such reactions is called ionizing.

Due to their energy, ionizing radiation is penetrating, that is to say, it can pass through matter. The penetrating power depends on the type of radiation and the stopping power of the material. This defines different thicknesses of materials to protect against them, if necessary and if possible.

### -2 Sources of ionizing radiation

Ionizing radiation has been present on Earth since its creation. Scientific progress has led men to use artificially produced ionizing radiation. Today, these radiations therefore have very diverse origins.

-Cosmic radiation is ionizing radiation of natural origin. They can come from the Sun. They are made up of atomic nuclei, high-energy particles and electromagnetic radiation. Their interaction in the atmosphere produces radioactive elements, said to be of cosmogenic origin, as well as pions which decay to produce muons.

Some electromagnetic radiation is also ionizing radiation. Conventionally, it is considered that electromagnetic radiation is ionizing at wavelengths below  $0.1 \mu\text{m}$ . Among the electromagnetic spectrum,



gamma rays, X-rays and certain ultraviolet rays are therefore considered ionizing. Gamma rays come from nuclear de-excitation following radioactive decay. X-rays and ultraviolet radiation come from electromagnetic processes such as electronic transition. They are part of cosmic radiation but are also produced artificially to be used in various fields such as scientific research, medical radiology or industry.

-Radioactivity produces different types of ionizing radiation:  $\alpha$  particles,  $\beta$  particles ( $\beta^-$ : electrons,  $\beta^+$ : positrons), protons, neutrons and  $\gamma$  rays. The radionuclides responsible for this radioactivity themselves have several origins:

Certain particle radiation is also considered ionizing radiation. They come from the various natural sources above but can also be directly created artificially and used in particle accelerators: electrons, protons, ions.

### **I-3 Modes of exposure to radiation**

Depending on how the radiation reaches the body, there are two modes of exposure: external or internal.

External exposure takes place when the subject is exposed to radiation sources external to them (radioactive substances in the form of clouds or deposits on the ground, sources for industrial or medical use, etc.). External exposure can affect the entire body or only part of it. It stops as soon as we are no longer in the path of the radiation (for example, in the case of a chest x-ray).

] Internal exposure (internal contamination) is possible when radioactive substances are inside the body. These cause internal irradiation. They may have entered by inhalation, by ingestion, through a wound or transcutaneously, and are then distributed throughout the body. We then speak of internal contamination. This only stops when the radioactive substances have disappeared from the body after a more or less long time by natural elimination, radioactive decay, and/or treatment.

Since 2006, regulations have defined several modes of exposure:

1. non-contact (remote) external exposure: irradiation;
2. external exposure with contact: external contamination;
3. internal exposure: internal contamination.

Contamination can be surface or volume (atmospheric).

### **I-4 Dosimetry of ionizing radiation:**

The first “dose” measurement was proposed by LIND in 1911, who observed that one gram of radium in the air produced 0.7 g of ozone per hour. The imprecision of the measurements remains exaggerated.

In the medical field, all interactions of ionizing radiation come down to a transfer of electronic kinetic energy. The idea of measuring an electrical charge created in the air by a beam of X-rays was attractive. It is in fact much easier to measure a charge created in a small mass of air than to assess the very small associated temperature difference, which is the reference method.

#### **I-4-1 Absorbed dose**

The physical quantity which makes it possible to quantify the interaction of radiation with matter is the absorbed dose.

If we consider radiation of incident energy  $E_i$  which enters an element of volume of mass  $dm$  and which emerges with an energy  $E_f$ , the absorbed dose is the ratio:

$$D = (E_i - E_f) / dm$$

The international unit of absorbed dose is the Gray or Gy. 1 Gy corresponds to the dose absorbed by a mass of 1 kg at which the radiation imparts an energy of 1 J, i.e.:  $1 \text{ Gy} = 1 \text{ J} \cdot \text{kg}^{-1}$ . The absorbed dose is measured with a dosimeter.

A historical unit is still used: the rad with  $1 \text{ Gy} = 100 \text{ rad}$ .

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Indicates the rate with which the absorbed dose increases.

$$D' = D / t \text{ (= absorbed dose / elapsed time)}$$

The international unit of absorbed dose is Gray/second (Gy/s), more practical:  $\mu\text{Gy/h}$

Also: Dose rate at a distance  $X = (\text{dose equivalent constant} \times \text{source activity}) / X^2$

However, exposures leading to the same absorbed dose value do not necessarily present the same risks, which are also linked to the nature of the radiation, the conditions of exposure and the sensitivity of the irradiated organs and tissues. Knowledge of the absorbed dose is therefore not sufficient to quantify the biological impact of exposure.

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A historical unit is still used: the rad with 1 Gy = 100 rad.

#### I-4-2 Equivalent dose

To assess the biological impact of radiation on an organ or tissue during uniform exposure, a so-called equivalent dose is calculated, which is the product of the absorbed dose by a dimensionless radiological weighting factor characteristic of the radiation. This dose is then expressed in sievert (Sv), named after the Swedish radiobiologist Rolf Sievert (1896 – 1966).

The equivalent dose is currently denoted HR, there comes the fundamental relationship:

$$H_R = D_R \cdot W_R$$

For a given radiation (R index), the equivalent dose is equal to the product of the absorbed dose and the quality factor.

The radiation quality factor  $W_R$

$W_R = 1$  for X-rays, gamma rays, beta rays

$W_R = 5$  for protons with energy greater than 2 MeV, for neutrons with energy greater than 20 MeV or less than 10 keV

$W_R = 10$  for neutrons with energy between 10 and 100 keV or between 2 and 20 MeV

$W_R = 20$  for neutrons with energies between 0.1 and 2 MeV and for alpha rays or heavier nuclei.

Unit: Sv/Gy

#### I-4-3 Effective dose

It is a quantity always expressed Sievert, determined by calculation:

$$H_E = \sum [W_T \times H_T]$$

= sum of all products between  $W_T$  tissue weighting factor and the  $H_T$  equivalent dose for tissue T.

The  $W_T$  tissue weighting factor

Different organs have different radiosensitivity (in terms of risk of cancer or hereditary conditions). To take this fact into account, and with the aim of defining a single scale on which risks can be quantified, regardless of whether the whole body or part of the body has been exposed, a  $W_T$  tissue weighting factor has been introduced which takes into account the different radiosensitivity of different organs.

Unit: / (without unit)

| Factor $W_T$ | Organes                                       |
|--------------|-----------------------------------------------|
| 0,2          | Gonades                                       |
| 0,12         | Colon, poumons, estomac, moelle osseuse rouge |
| 0,05         | Seins, vessie, foie, œsophage, thyroïde       |
| 0,01         | Os (surface, peau)                            |
| 0,05         | Rest of the body                              |

#### Meaning of effective dose

The indication of the effective dose is synonymous with the indication of RISK.

For the same effective dose (e.g. 10 mSv), the risk due to radiation is independent of whether a person has suffered partial exposure (e.g. of the skull, neck and thorax) or whole body exposure.

The definition of radiation risk includes

- \* the risk of dying from radiation-induced cancer
- \* the risk of developing radiation-induced cancer
- \* the risk of producing genetic effects across all generations.

In the formula for calculating the effective dose, the relative sensitivity of different organs to produce one of the above effects (tissue weighting factor), the type and energy of the radiation and the distribution of absorbed dose are taken into account.

The effective dose is a quantity determined by calculation and which takes into account:

- the type and energy of the radiation
- whether there has been external or internal exposure
- whether a part or the whole body has been exposed

The risk of fatal cancer is estimated at 5%/Sv and that of hereditary risks at 1%/Sv at low doses.

#### **I-5 Biological effects of ionizing radiation**

These are the effects observed when ionizing radiation interacts with living tissue by transferring their energy to organic molecules. The severity of these effects depends on the type of radiation (alpha, beta, gamma), the absorbed dose, but also the absorption rate and the radio sensitivity of the tissues concerned. The biological effects of rapid irradiation are very different from those of long irradiation. The first cause cell death and appear within a few hours, days or weeks. The others are better tolerated because part of the lesions are repaired. However, doses of radiation that are too low to destroy cells can nevertheless cause cellular changes, the consequences of which appear after several years.

Each biological cell contains approximately  $10^{14}$  molecules divided into  $10^{13}$  water molecules and  $10^9$  large molecules whose importance for cellular life varies from one molecular species to another hierarchically: (DNA; RNA, enzymes, proteins)

The effects of ionizing radiation on an organism result from the transfer of energy to matter, a transfer which constitutes a complex process.

##### **I-5-1 At the molecular level (Molecular lesions)**

→ Lesions induced by direct energy transfer

The energy from the radiation is transferred directly to the molecule which is ionized or excited.

The excess energy acquired by the molecule is lost by:

- Fluorescence radiation;
- breakage of chemical bonds.

Example: the excess energy of an ionization is 12.6 eV and that of an excitation is 6.6 eV. In both cases, the excess is sufficient to break the H-OH bond of a water molecule whose energy is 5.16 eV.

→ Lesions induced by indirect energy transfer

The energy is transferred to molecules in the vicinity of the molecules concerned which in turn induce damage to the molecules concerned. This is typically the action of free radicals formed during the radiolysis of water. Currently, it is believed that the direct effect predominates.

Indeed, due to their short lifespan (10<sup>-6</sup> seconds) free radicals migrate at short distances. As a result, only those formed near a molecule can harm it.

→ The targets of radiation-induced lesions

All biological molecules are targets. We can mention in particular:

has. water: due to its proportion in all living organisms (around 70% in humans)

b. DNA: due to its central function in the cell (alteration of chromosomes and cell division, alteration of genes and mutations).

##### **I-5-2 At the cellular level (Cellular damage)**

→ Cell death, carcinogenesis and hereditary effects

Cell death results from irreparable damage to structures vital to the cell such as chromosomes. Cell death can be immediate for high doses (several Gy) or delayed for low doses.

The effects of low doses are as follows:

- the lesions are in most cases repaired and the cell returns to normal
- if the lesions are poorly repaired, the cells can lose their capacity to divide from the 1st division (1st mitosis) or after several mitoses (delayed death)

Finally, the cell may appear normal in terms of survival, but be abnormal in chromosomal terms and suffer genetic (hereditary) effects or move towards the development of cancers.

##### **I-5-3 The pathological effects of ionizing radiation in humans**

During exposure, if the number of cells destroyed is too great, their replacement by the multiplication of viable cells is insufficient. The resulting cellular deficit leads to tissue damage which in turn impairs organ function. Finally, this can lead to clinical signs or even death of the individual.

Deterministic effects are therefore those that we observe beyond a threshold, which corresponds to a dose of 0.2 to 0.3 Gy. They are called deterministic because they are obligatory, that is to say they always appear.

### Stochastic effects

Stochastic effects are the long-term probabilistic consequences, in an individual or in its descendants, of the transformation of a cell.

They therefore result from poorly repaired lesions of DNA molecules.

Stochastic effects are of 2 types:

- if the mutation concerns a gene of a somatic cell, the effects concern the exposed individual. These are the carcinogenic effects
- if the mutation concerns a gene of a germ cell, the effects concern the exposed individual but can also be transmitted. These are hereditary effects.

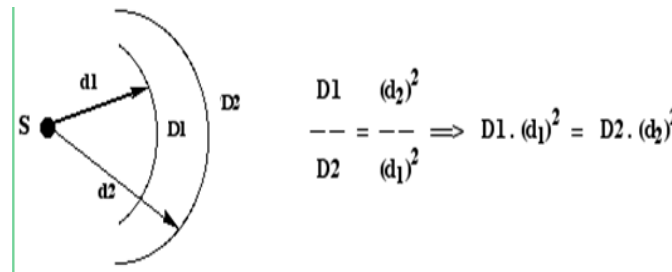
## I-6 PROTECTION AGAINST IONIZING RADIATION

The effects on the body of radiation exposure depend on the nature of the radiation:

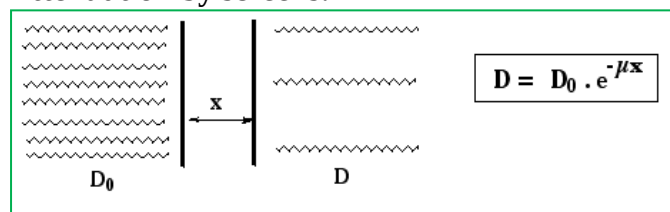
The absorbed dose received by a person is the product of the exposure time by the rate of the absorbed dose.

So protection against external exposure is done either by:

1. exposure time.
  2. Minimization of absorbed dose rate.
- Minimizing the distance to the source: Find a way to increase the distance between the operator and the radiation source



### Attenuation by screens:



## II- NON-IONIZING RADIATION

Non-ionizing radiation refers to a type of radiation for which the electromagnetic energy carried by each quantum is insufficient to cause the ionization of atoms or molecules. These radiations can, however, have enough energy to cause an electron to move to a higher energy level. Some of this radiation can have biological effects.

Non-ionizing radiation includes near ultraviolet radiation, visible light, infrared, microwaves, radio waves and static fields.

Among the radiation cited as non-ionizing, the most energetic (near ultraviolet, visible light) can in certain cases ionize a few molecules.

Sunlight, largely filtered by the Earth's atmosphere, arrives at the Earth's surface essentially composed of non-ionizing radiation (with the notable exception of certain ultraviolet radiation).

### Biological effects of non-ionizing radiation

|               | Source                            | Wavelength   | Frequency | Biological effect                           |
|---------------|-----------------------------------|--------------|-----------|---------------------------------------------|
| Ultraviolet C | Light for radiation sterilization | 100 - 280 nm |           | Erythema, skin pigmentation, photokeratitis |

|                        |                                                                                                  |                         |                   |                                                                                                                      |
|------------------------|--------------------------------------------------------------------------------------------------|-------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------|
|                        |                                                                                                  |                         |                   |                                                                                                                      |
| <u>Ultraviolet B</u>   | Tanning booth                                                                                    | 280 - 315 nm            |                   | photokeratitis, erythema, skin pigmentation, skin cancer, photosensitive reactions of the skin, vitamin D production |
| <u>Ultraviolet A</u>   | black light, sunlight                                                                            | 315 - 400 nm            |                   | Photochemical cataract, erythema, skin pigmentation                                                                  |
| <u>Lumière visible</u> | Lasers, sunlight                                                                                 | 400 - 780 nm            |                   | Skin aging, skin cancer, photochemical and thermal retinal damage.                                                   |
| <u>Infrarouge A</u>    | Lasers, remote controls                                                                          | 780 nm - 1,4 $\mu$ m    |                   | Thermal retinal burn, thermal cataract, sunburn                                                                      |
| <u>Infrarouge B</u>    | Lasers, long distance communications                                                             | 1,4 $\mu$ m - 3 $\mu$ m | 215 THz - 100 THz | Corneal burn, cataract, sunburn                                                                                      |
| <u>Infrarouge C</u>    | laser <u>Infra-rouge</u>                                                                         | 3 $\mu$ m - 1 mm        | 100 THz - 300 GHz | Brûlures à la <u>cornée</u> , <u>cataracte</u> , échauffement de la surface du corps                                 |
| <u>Micro-ondes</u>     | Mobile phones in the PCS range, microwave ovens, cordless phones, motion detectors, radar, Wi-Fi | 33 cm - 1 mm            | 1 GHz - 300 GHz   | Heating of body tissues                                                                                              |
| <u>Ondes radio</u>     | Cell phones, television, radio transmitters                                                      | 3 km - 33 cm            | 100 kHz - 1 GHz   | Heating of the human body to a thickness of up to 1 cm                                                               |
| Low frequency          | electricity transmission cables                                                                  | > 3 km                  | < 100 kHz         | Accumulation of electrical charges on the surface of the body, disruption of the response of muscles and nerves      |
| Static field           | strong magnets, MRI                                                                              | infinite                | 0 Hz              | Dizziness, nausea, electrical charges on the surface of the body                                                     |

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**Dr. C. SEKKAK**

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